Analysing Adversarial Attacks on U-Net Models for Inland Water Segmentation from SAR Images

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***Abstract*—**
The Segmentation of Water Bodies from satellite images is an important task which is used for numerous other larger and more complicated tasks such as Flood Mapping, and Disaster Management. While comprehensive satellite imaging systems are available to obtain high resolution images of various river basins, the corresponding water masks are not available. Manual annotation of the images is prone to a variety of errors. In this work, we automate the process of mask generation using Sentinel 2 images, and simulate manual errors that a human may cause, in the form of adversarial attacks on the model, and observe the performance of U-Net on these corrupted images, and study the robustness of the model to human errors in annotation.

***Index Terms*—Deep learning, SAR Images, Image Segmentation, U-Net, Adversarial Attacks**

I. Introduction

WATER body segmentation is a task which has numerous applications in a number of fields, ranging from Flood Mapping and Monitoring [1], Glacial Lake Segmentation [2] and studying Sedimentation and Erosion [3]. Most Segmentation approaches are based on the applications of Convolutional Neural Networks (CNNs), primarily using encoder-decoder approaches such as U-Net [4].

There exist a number of approaches for the water body segmentation task. Thresholding exists as a primary approach for segmentation, but the thresholds are identified experimentally based on a subset of data [5]. This lacks the capability to generalise, and thresholds may be different for different subsets. In [6], the authors propose a tool which utilises spectral indices for river identification. The authors in [7] proposed a mathematical morphological based approach for identifying rivers. However, determining the parameters of path opening and closing is a challenging task and can affect the results obtained. Other approaches include [8], which performs segmentation on Sentinel-1 SAR imagery, similar to our work, but uses Unsupervised Learning based methods.

Semantic Segmentation primarily uses deep neural networks to segment the pixels in an image into various classes. Several approaches have used CNNs for semantic segmentation of water bodies. [1] uses networks including fcn-8s [9], tiramisu [10], and pix2pix [11], which is a GAN which has been adapted for the segmentation task. The labels for the dataset in this work has been created by a human annotator.

Other interesting works include a Dynamic Snake Convolution [12], which can be used for Segmentation of Tubular structures such as blood vessels and roads, and may be adapted for river segmentation. The work by [13] analyses a number of traditional and deep learning based approaches for semantic segmentation of water bodies.

Our work for semantic segmentation is primarily inspired by [14], which uses two primary segmentation models, namely U-Net [4] and DeepLabV3+ [15] for segmentation of Sentinel-1 SAR images. Our work draws inspiration from this work.

While there exist a number of channels that allow us to easily obtain satellite images, including Sentinel-1 SAR imagery [16] provided by the European Space Agency (ESA), a primary concern is to obtain datasets which are well suited to the task and are highly accurate. Several efforts have been made to curate data sets that are suitable for a variety of tasks. Examples include the *Sen1Floods11* dataset [17], which uses raw Sentinel-1 imagery, and classifies flood water and permanent water, for flood mapping at a global scale, and the Hydrosheds Dataset [18] which provides hydrographic data products including catchment boundaries, rivers, lakes etc. Other examples include DIVA-GIS [19], which provides comprehensive shapefiles for river basins which can be used for ground truth generation, and MERIT Hydro [20], which provides high-resolution hydrography maps using elevation data and water body datasets such as G1WBM (Global 1-second Water Body Map), GSWO (Global Surface Water Occurence), and OpenStreetMap.

However, we find that these datasets are not particularly suited for our task. We find that the S1-S2 Water Dataset [21] is the best suited for the semantic segmentation task, and we adopt their methodology for dataset generation in our work. This work uses the NDWI index calculated from Sentinel-2 images to create water masks, and we find that this methodology is highly accurate for our work.

Many of the works stated use manual annotation to generate the masks to be used as ground truths. These are prone to a number of human errors, such as change of the river width, and mislabeling boundary pixels. This motivates our investigation into error types which can be introduced by manual annotation, and simulation of those errors while analysing the performance of segmentation architectures. This leads us to the usage

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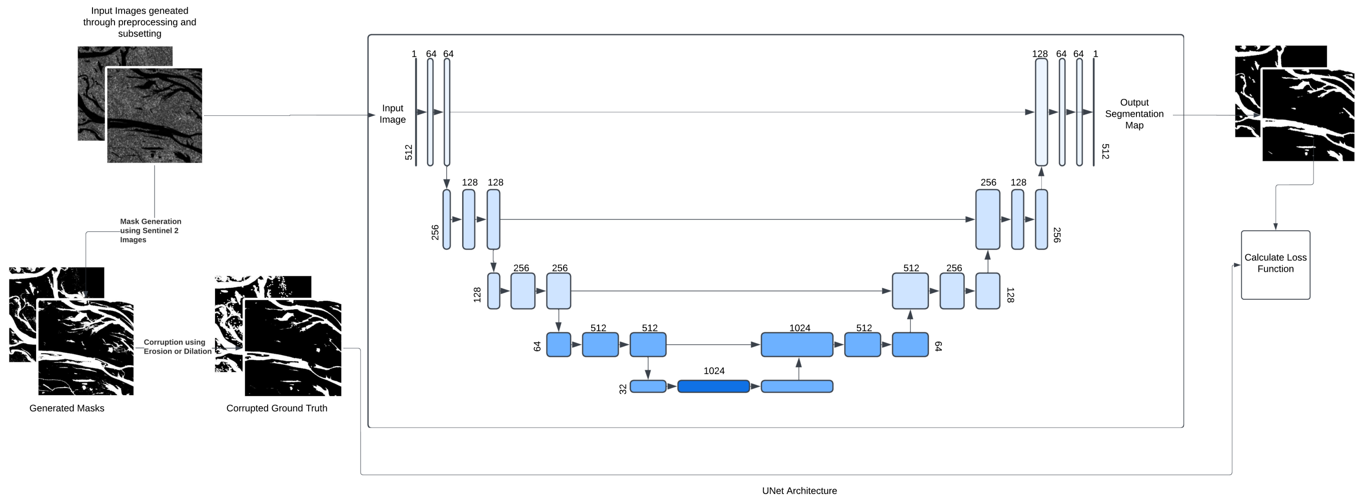


Fig. 1: Overview of the procedure used for Adversarial Testing for Water Body Segmentation

of morphological operations, and adopting an adversarial learning-based approach. We adopt a corruption strategy which utilizes morphological operations in a controlled manner to generate modified ground truth masks.

In the field of adversarial attacks and defenses, significant research has been conducted to safeguard deep neural networks (DNNs) from adversarial manipulations that can degrade their performance. One notable work is the introduction of Vector Quantization U-Net (VQUNet) [22], which addresses the vulnerability of DNNs by employing a novel noise-reduction mechanism to combat adversarial attacks. VQUNet leverages a discrete latent representation learning through a multi-scale hierarchical structure to both reduce adversarial noise and reconstruct the original data with high fidelity. Gu et al. (2021) [23] present a domain adaptation approach for semantic segmentation by leveraging the NDWI index to enhance the edge detail in pixel-level classifications. The use of NDWI improves segmentation accuracy by addressing data distribution differences between source and target domains. Their adversarial learning framework aligns segmentation outputs, which bears similarity to adversarial attacks in NLP and vision, where data discrepancies are exploited to compromise model robustness and consistency across domains.

Our major contributions include -

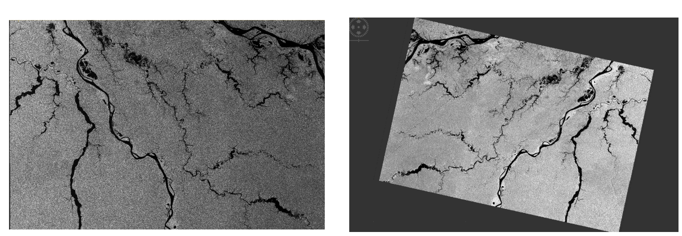
1) A method to generate accurate ground truth labels for water body segmentation, using Sentinel 2 images and NDWI index.
2) A methodology for introducing human reproducible errors into the binary masks, through the use of morphological operations (erosion and dilation, described later).
3) Testing and analysis of the UNet architecture under various levels of corruption, to observe the limits of high performance, beyond which the model undergoes degradation.
4) A qualitative observation and explanation of the results on a few data points.

II. Method

A. Image Acquisiton and Preprocessing

We selected Sentinel-1 SAR data for the Amazon River Basin from platforms like the Sentinels Scientific Data Hub and Copernicus Open Access Hub. We focus on this basin primarily due to the rich variety of river structures observed in the region, which can help in robust training.

Sentinel-1 GRD products are obtained using the Alaska Satellite Facility (ASF) API, following which we apply a number of preprocessing steps. The first step is Orbit File Application [24]. The orbit file updates the original metadata of SAR product. Spectral data acquired by satellite sensors is influenced by a number of factors, such as atmospheric absorption, scattering and sensor-target-illumination geometry. Radiometric calibration [25] ensures that normalized images would appear as if they were acquired with the same sensor under similar atmospheric and illumination conditions. Speckle filtering [26] is used to create a denoised image with enhanced amplitude and suppressed noise. Finally, we perform Terrain correction [27], [28], which allows us to correct geometric distortions by moving image pixels into proper spatial relationship with each other based on a Digital Elevation Model (DEM). Figure 2 shows the result of applying these preprocessing techniques on image quality.



(a) Original Image (b) Preprocessed Image

Fig. 2: Original vs Preprocessed Image. This highlights the importance of the various preprocessing steps taken.

To augment the dataset, we extracted smaller subsets from the Sentinel-1 products, which can be easily managed and used. This entire process resulted in a total of 1263 images, which we use for this work. The generation of binary water masks is detailed in the next subsection.

B. Water Mask Generation

Initial approaches for dataset generation involved the use of predefined shapefiles, which contain water networks, which can be used to obtain the required mask. We experimented with a number of shapefiles, including Diva GIS [19], Merit HYDRO [20] and Hydrosheds [18], but none of them generated accurate enough water masks which were particularly suited to the segmentation task.

Finally, we utilized an approach similar to that used by the authors of [21]. Multispectral Sentinel 2 images were procured using Google Earth Engine, and the regions corresponding to the latitude and longitude for each image was clipped out. For each image, we calculate the NDWI index of the image. This index is used to identify the water regions from the land regions, using a suitable threshold. This resulted in highly accurate water masks. An example is shown in Figure 3.

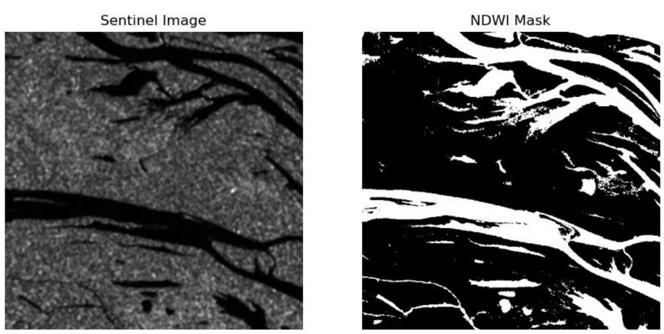


Fig. 3: Image along with the Generated Mask

C. Adversarial Experiments

We began by training a baseline model using a vanilla U-Net architecture, using the sentinel 1 imagery, and generated water masks as ground truth annotations. Our goal with this initial setup was to establish a performance benchmark. We utilise a train-val-test split of 70-10-20, and train the model for 20 epochs, across all the experiments.

Following this, we introduce controlled corruptions into the masks by using erosion and dilation, as these operations present examples of human introducible errors.

Erosion and Dilation are two morphological operations in Image Processing which we use to bring about a controlled form of corruption, which a human may introduce during manual annotation. In a morphological operation, the value of each pixel in the output image is based on a comparison of the corresponding pixel in the input image with its neighbours. The number of neighbours is decided by the kernel size.

In Erosion, the value of the output pixel is the minimum value of all pixels in the neighborhood. In a binary image, a pixel is set to 0 if any of the neighboring pixels have value 0. This simulates scenarios where features might be diminished.

In Dilation, the value of the output pixel is the maximum value of all pixels in the neighborhood. In a binary image, a pixel is set to 1 if any of the neighbouring pixels have the value 1. This simulates scenarios where features might be enlarged.

The methodology for corruption is as follows - out of the train and the val images, we pick out one-third of the images for corruption using erosion, one-third using dilation, and one-third is left uncorrupted. For an experiment, we then set a threshold on the number of corrupted pixels allowed.

Erosion and Dilation are applied to the corresponding masks iteratively, till either 100 iterations are over, or the number of corrupted pixels crosses the threshold, following which we choose the previous highest corrupted version of the mask.

It is to be noted that although we set a limit on the number of corrupted pixels, in practice, we may not be able to reach that limit. Erosion results in the decrease of the number of white pixels, and hence if not many white pixels are present in the first place, the number of corrupted pixels will not increase.

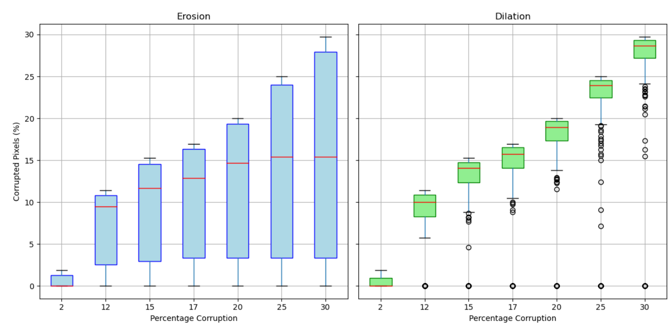


Fig. 4: Boxplots showing the actual corruption versus the threshold

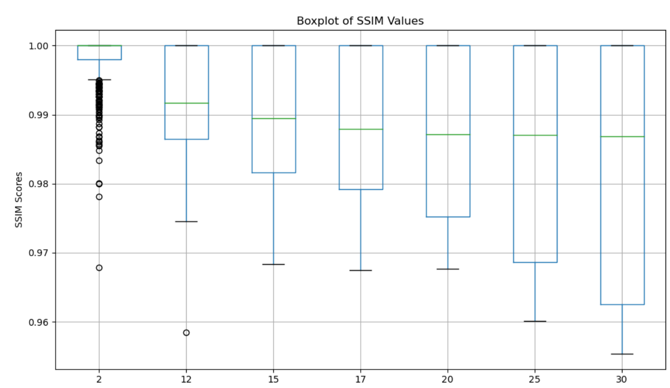


Fig. 5: Boxplots showing the SSIM values between the actual mask and the corrupted masks for various levels of corruption

This is demonstrated in Figures 4 and 5. We observe that the number of pixels corrupted for images undergoing erosion becomes stagnant as we increase the corruption levels, with the median and the lower quartile barely changing beyond 20% corruption. The dilation plots, on the other hand, show proper trends. The same is true for the Structural Similarity

TABLE I: Performance Metrics of U-Net Model at Various Corruption Levels

Metric	Clean	2%	12%	15%	17%	20%	25%	30%
Average Dice Coefficient	0.7696	0.7748	0.7693	0.7678	0.7580	0.7252	0.5971	0.2532
Average IoU	0.8230	0.8252	0.8224	0.8190	0.8141	0.7897	0.7081	0.4964
Average Precision	0.8959	0.8959	0.8989	0.8899	0.9028	0.9024	0.8746	0.8178
Average Recall	0.8688	0.8734	0.8641	0.8663	0.8536	0.8303	0.7592	0.5878
Average F1 Score	0.8652	0.8677	0.8646	0.8633	0.8591	0.8405	0.7678	0.5672
Average Specificity	0.9708	0.9680	0.9748	0.9705	0.9785	0.9812	0.9842	0.9963
Model's Average Pixel Accuracy	0.9554	0.9546	0.9546	0.9541	0.9523	0.9460	0.9196	0.8282

Index (SSIM) value between the actual masks (as calculated using the NDWI index), and the corrupted masks. Hence the corruptions may be limited by the images themselves.

We also introduce randomness in the kernel sizes used for corruption, to allow for some non-uniformity in the corruption. The stochastic process of selecting the kernels is essential when we need the model to not be biased in its learning. The chosen size depends on the white-pixel ratio of the image, with a larger kernel size for erosion being preferred for images with less white pixel ratio, and vice versa for dilation. Finally, we utilize the following levels of corruption (the image sizes are 512×512) - 2%, 12%, 15%, 17%, 20%, 25%, and 30%.

III. Results

Table I shows the evaluation metrics of training the UNet model, across the various levels of corruption. We get a good 95 % accuracy in segmentation. Furthermore, the performance does not seem to diminish by a large amount even after increasing the corruption levels to as high as 20%. This is surprising, as the model is able to tolerate even such high levels of corruption. We observe that the model performance does not diminish significantly till around 25% corruption. Hence, we investigate more into the training of the model, the qualitative predictions of the model on a particular image, and the actual corruption that the model is facing.

Figure 6 shows the per epoch accuracy on the training and validation data for each of the corruption levels. The plots clearly indicate that until 17% corruption, the validation accuracy never falls significantly, beyond which the model begins to show instability in the validation accuracies. Further, we observe that the model is able to avoid overfitting, as with the increase in the corruption level, the training accuracy falls, but never the validation accuracy.

These results are further supported by Figure 7. The figure shows the input image, and the corresponding actual masks which are fed to the model as inputs. This is a point on which erosion has been applied, and we observe that with the increase in the corruption levels, the percentage of white pixels reduces significantly, with no remaining white pixels when the corruption is increased to 25%. This is an example of a data point where just increasing the corruption level does not lead to an increase in the actual corrupted pixels.

We observe that the UNet output before applying the threshold becomes increasingly blurred with the increase in corruption, indicating the model's difficulty in discerning land from water. However we still observe that the model is able to find the river structure, and when we apply the threshold on the UNet output, the masks predicted are still reasonably accurate, which suggests UNet's ability to be robust to high levels of corruption.

All these results show that the model is able to easily tolerate 17% corruption, beyond which the model becomes instable in training. The model avoids overfitting and, despite the high levels of corruption, is able to accurately obtain the river structure. Till 20% corruption, the model is even able to accurately find the mask after thresholding, beyond which the predictions become incresingly blurred and poor.

IV. Conclusions and Future Works

In this work, we introduced a new method to simulate human introduced errors using morphological operations, and used a controlled corruption strategy to test the UNet model's performance. Through testing and the observed results, It is clear that the model is robust to morphological forms of corruption, and is able to detect water structures accurately even in the presence of high corruption levels, to a certain corruption threshold. An interesting line of work would be to observe other, more traditional modes of corruption, including rotation and translation, and observe their effects.

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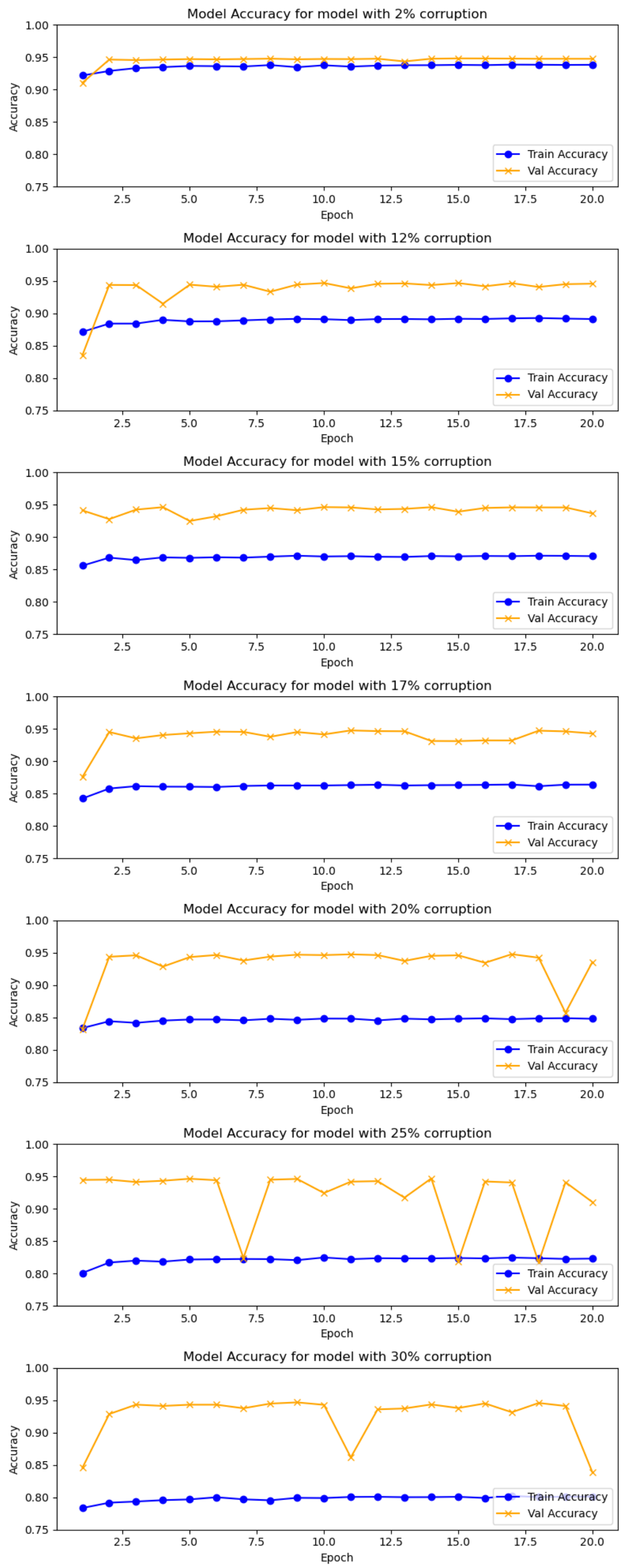

Fig. 6: Training and Validation Accuracy Plots

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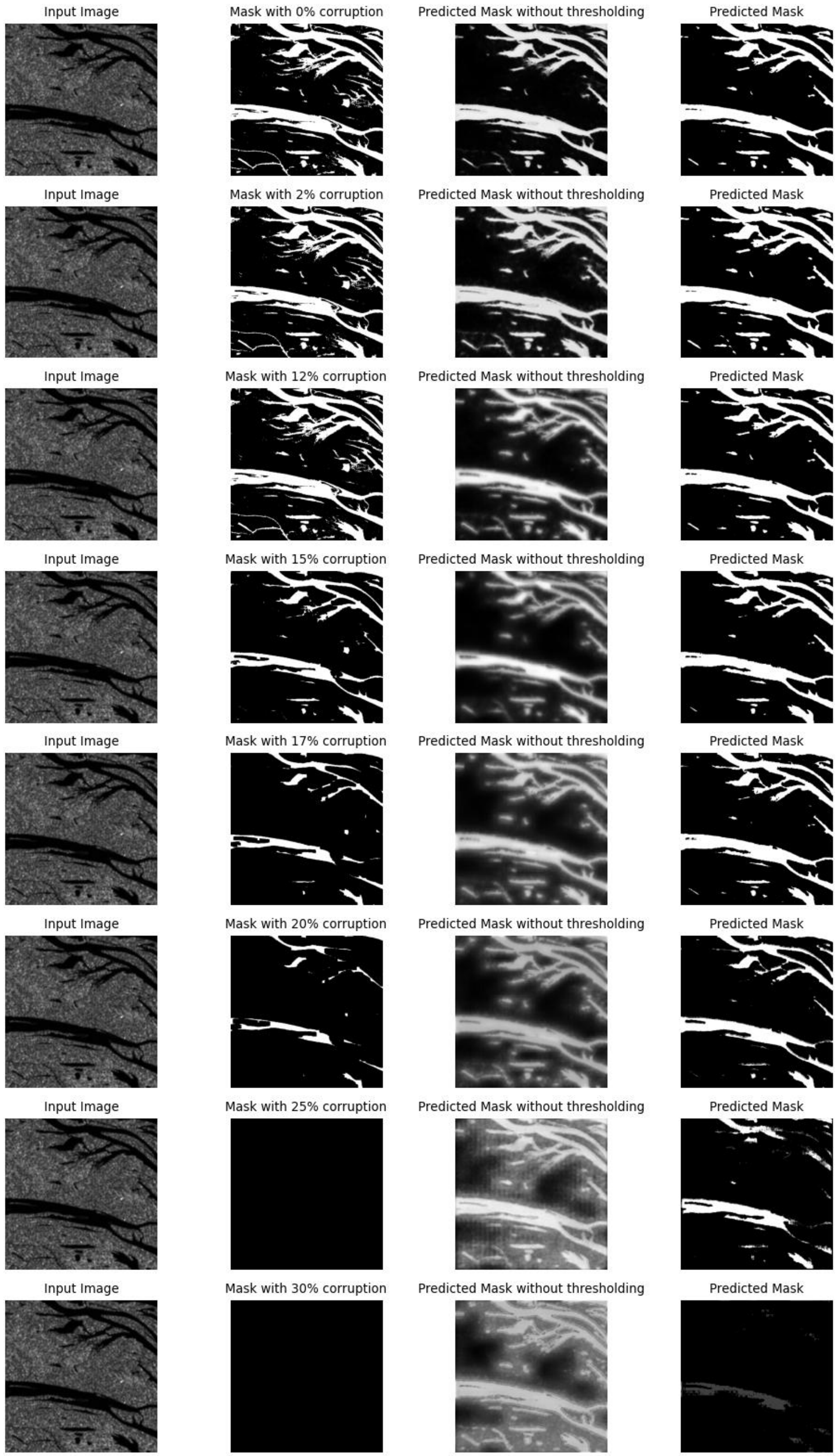

Fig. 7: Qualitative Plot showing the effect of corruption level on the predicted and actual masks for one datapoint